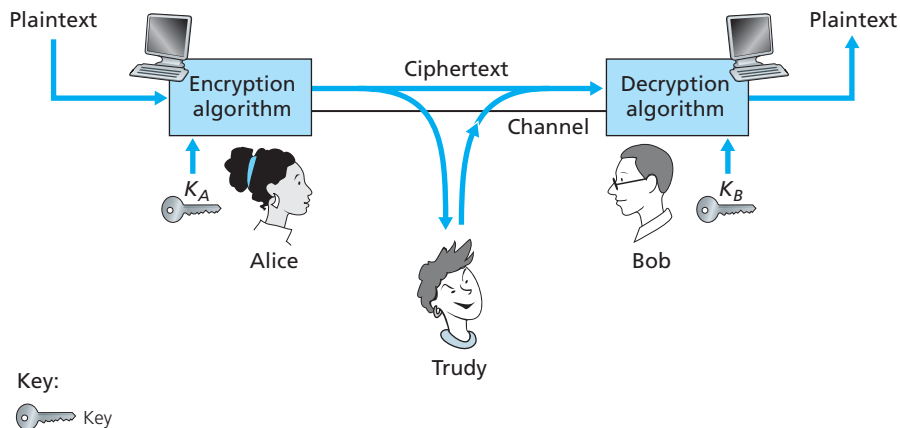


## 8.2 Principles of Cryptography

Although cryptography has a long history dating back at least as far as Julius Caesar, modern cryptographic techniques, including many of those used in the Internet, are based on advances made in the past 30 years. Kahn's book, *The Codebreakers* [Kahn 1967], and Singh's book, *The Code Book: The Science of Secrecy from Ancient Egypt to Quantum Cryptography* [Singh 1999], provide a fascinating look at the long history of cryptography. A complete discussion of cryptography itself requires a complete book [Kaufman 1995; Schneier 1995] and so we only touch on the essential aspects of cryptography, particularly as they are practiced on the Internet. We also note that while our focus in this section will be on the use of cryptography for confidentiality, we'll see shortly that cryptographic techniques are inextricably woven into authentication, message integrity, nonrepudiation, and more.

Cryptographic techniques allow a sender to disguise data so that an intruder can gain no information from the intercepted data. The receiver, of course, must be able to recover the original data from the disguised data. Figure 8.2 illustrates some of the important terminology.

Suppose now that Alice wants to send a message to Bob. Alice's message in its original form (for example, "Bob, I love you. Alice") is known as **plaintext**, or **cleartext**. Alice encrypts her plaintext message using an **encryption algorithm** so that the encrypted message, known as **ciphertext**, looks unintelligible to any intruder. Interestingly, in many modern cryptographic systems, including those used in the Internet, the encryption technique itself is *known*—published, standardized, and available to everyone (for example, [RFC 1321; RFC 3447; RFC



**Figure 8.2** ♦ Cryptographic components

2420; NIST 2001]), even a potential intruder! Clearly, if everyone knows the method for encoding data, then there must be some secret information that prevents an intruder from decrypting the transmitted data. This is where keys come in.

In Figure 8.2, Alice provides a **key**,  $K_A$ , a string of numbers or characters, as input to the encryption algorithm. The encryption algorithm takes the key and the plaintext message,  $m$ , as input and produces ciphertext as output. The notation  $K_A(m)$  refers to the ciphertext form (encrypted using the key  $K_A$ ) of the plaintext message,  $m$ . The actual encryption algorithm that uses key  $K_A$  will be evident from the context. Similarly, Bob will provide a key,  $K_B$ , to the **decryption algorithm** that takes the ciphertext and Bob's key as input and produces the original plaintext as output. That is, if Bob receives an encrypted message  $K_A(m)$ , he decrypts it by computing  $K_B(K_A(m)) = m$ . In **symmetric key systems**, Alice's and Bob's keys are identical and are secret. In **public key systems**, a pair of keys is used. One of the keys is known to both Bob and Alice (indeed, it is known to the whole world). The other key is known only by either Bob or Alice (but not both). In the following two subsections, we consider symmetric key and public key systems in more detail.

### 8.2.1 Symmetric Key Cryptography

All cryptographic algorithms involve substituting one thing for another, for example, taking a piece of plaintext and then computing and substituting the appropriate ciphertext to create the encrypted message. Before studying a modern key-based cryptographic system, let us first get our feet wet by studying a very old, very simple symmetric key algorithm attributed to Julius Caesar, known as the **Caesar cipher** (a cipher is a method for encrypting data).

For English text, the Caesar cipher would work by taking each letter in the plaintext message and substituting the letter that is  $k$  letters later (allowing wrap-around; that is, having the letter  $z$  followed by the letter  $a$ ) in the alphabet. For example if  $k = 3$ , then the letter  $a$  in plaintext becomes  $d$  in ciphertext;  $b$  in plaintext becomes  $e$  in ciphertext, and so on. Here, the value of  $k$  serves as the key. As an example, the plaintext message “bob, i love you. alice” becomes “ere, l oryh brx. dolfh” in ciphertext. While the ciphertext does indeed look like gibberish, it wouldn't take long to break the code if you knew that the Caesar cipher was being used, as there are only 25 possible key values.

An improvement on the Caesar cipher is the **monoalphabetic cipher**, which also substitutes one letter of the alphabet with another letter of the alphabet. However, rather than substituting according to a regular pattern (for example, substitution with an offset of  $k$  for all letters), any letter can be substituted for any other letter, as long as each letter has a unique substitute letter, and vice versa. The substitution rule in Figure 8.3 shows one possible rule for encoding plaintext.

The plaintext message “bob, i love you. alice” becomes “nkn, s gktc wky. mgsbc.” Thus, as in the case of the Caesar cipher, this looks like